

PRASENJIT KARMAKAR

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[Website](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Professional Summary

PhD researcher at IIT Kharagpur (Prime Minister's Research Fellow) with hands-on experience in generative modeling, multimodal AI, and LLM and VLM systems, including a visiting research role at Singapore Management University focused on embodied AI and LLM task planning. Experienced in owning end-to-end machine learning pipelines, from data collection and model design through deployment. Author of publications at ACM CHI, NeurIPS, and IEEE PerCom, and inventor on four filed patents.

Technical Skills

Programming Languages and Concepts: C/C++, Java, Python, Object Oriented Programming

Artificial Intelligence: Data Analytics, Machine Learning, Deep Learning, Reinforcement Learning, model quantization and deployment, MLOps, Model drift management

NLP and Generative AI: LLM and VLM systems, Generative AI, model finetuning, embodied AI task planning, LLM guided system design, multimodal sensor and language fusion, KV-Caching, Inference Optimization

Libraries and Frameworks: Scikit-Learn, Numpy, Pandas, Matplotlib, Scipy, Seaborn, PyTorch, TensorFlow, TFLite, Slurm, Unsloth, Hugging Face Transformer, PEFT, Transformer Reinforcement Learning

Applied Systems: IoT and sensor pipeline design, AR/VR and wearable interaction design, distributed systems, embedded systems, prototype design, 3D printing, PCB design, field studies and user studies

Research and Communication: Peer-reviewed publications, patent drafting, technical writing, teaching

Work Experience

Visiting Researcher, Singapore Management University, Singapore

July 2025 to January 2026

- Collaborated on Embodied AI and LLM task planning projects.
- Skills applied: Multimodal AI, LLM/VLM, Generative AI, Finetuning, Embodied AI.

Systems Engineer, Tata Consultancy Services, Bangalore, India

November 2020 to April 2022

- Worked as a Frontend Web Developer in the BFSI sector.
- Skills applied: Lit Element, React, Flask, Spring Boot.

Research Intern, National Institute of Technology Durgapur, Durgapur, India

September 2019 to September 2020

- Worked as an ML researcher developing smart city applications: Intelligent Bus Transport (BuStop) and City scale Air Quality Monitor (AQuaMoHo).
- Skills applied: Embedded Systems Programming, Python, Data Science, MLOps.

Education

PhD in Computer Science and Engineering (in progress), IIT Kharagpur, Kharagpur, India

July 2022 to December 2026 (Expected)

CGPA: 8.95. Research area: Ubiquitous Computing, PhD Advisor: Dr. Sandip Chakraborty.

B.Tech in Computer Science and Engineering, MAKAUT, Kolkata, India

July 2016 to July 2020

CGPA: 9.13. Thesis: Reinforcement Learning in Continuous Multi joint Environments.

Filed Patents

- An End to End System for Multi modal Physiological and Environmental Exposure Monitoring. India Patent Application 202631101632. Filed August 2026.
- An End to End System for Modulating Ambient Lights with Environmental Sensors and LLM Guided Co Learning. India Patent Application 202631099342. Filed August 2026.
- A System with Framework Architecture for Distributed Air Quality Monitoring in Indoor Spaces. India Patent Application 202531006180. Filed January 2025 (Published).
- A Framework for Monitoring and Ventilating Personal CO2 Bubbles through a Wrist Wearable and Virtual Reality Application. India Patent Application 202531001296. Filed January 2025 (Published).

Selected Publications

- Karmakar, P., Yadav, M., Rout, S., Pradhan, S., Chakraborty, S. From Invisible to Actionable: Augmented Reality Interactions with Indoor CO2. ACM CHI 2026 (top-tier conference).
- Karmakar, P., Pradhan, S., Chakraborty, S. Indoor Air Quality Dataset with Activities of Daily Living in Low to Middle Income Communities. NeurIPS 2024 Datasets and Benchmarks (top-tier conference).
- Sen, A., Mandal, A., Karmakar, P., Das, A., Chakraborty, S. mmDrive: mmWave Sensing for Live Monitoring and On Device Inference of Dangerous Driving. IEEE PerCom 2023 (top-tier conference).
- Karmakar, P., Pradhan, S., Chakraborty, S. Exploring Indoor Air Quality Dynamics in Developing Nations: A Perspective from India. ACM Journal on Computing and Sustainable Societies 2024 (Google's SCAI Award).
- Karmakar, P., Shah, V.K., Roy, S., Hazra, K., Saha, S., Nandi, S. Reliable Backhauling in Aerial Communication Networks Against UAV Failures: A Deep Reinforcement Learning Approach. IEEE Transactions on Network and Service Management 2022 (Q1 Journal)

Full list of papers available on Google Scholar.

Honors, Awards and Recognition

- Prime Minister's Research Fellowship, March 2023.
- Google Award for Inclusion Research, USD 35,000, October 2023.
- Google Award for Society Centered AI, USD 100,000, September 2025.
- LRN International Travel Grant and ACM IARCS Travel Grant, April 2026.
- COMSNETS Travel Grant, January 2025; ACM COMPASS Travel Grant, July 2024.

References will be provided on request.